

Pranay Harsha

0426076232 | pranayhk06@gmail.com | linkedin.com/in/phk04

PROFILE

Graduate Mechatronics Engineer with hands-on experience in control systems, embedded systems programming, and sensor integration. Proven ability to configure, test, and document technical systems in a professional engineering environment. Strong analytical and problem-solving skills with experience in automated systems, technical documentation, and cross-disciplinary collaboration. Seeking to apply hardware, software, and systems knowledge in a building automation environment.

EDUCATION

Macquarie University
BEng (Hons) Mechatronics

Sydney, New South Wales
Feb 2022 – Jun 2026

WORK EXPERIENCE

MQ Translational Orthopedic Research Lab
Research Intern – Cadaveric Knee Simulator

Sydney, New South Wales
Jul 2025 – Jun 2026

- Redesigning the cadaveric knee simulator's actuator system to increase stroke length and output force, applying systematic component evaluation, load analysis, and design review processes.
- Performing structural assessment of the existing rail mechanism to verify compatibility with higher-force actuators, ensuring mechanical integrity under increased loading conditions.
- Configured and tested system components, verifying performance against design specifications and documenting results for engineering review.
- Engineered a standardised workflow for specimen segmentation in (Dragonfly ors) and CAD reconstruction(Fusion360), reducing processing time by over 80% (16 → 3 hrs) through systematic process optimisation.
- Produced detailed technical documentation including SOPs and CAD records, maintaining accurate as-built records throughout the project lifecycle.

PROJECT EXPERIENCE

Macquarie University
Student engineer – Blade Runner

Sydney, New South Wales
Jul 2024 – Nov 2024

- Worked with a cross-functional team to design a functional monorail system with collision detection and Wi-Fi capabilities, integrating mechanical and electrical subsystems.
- Responsible for technical documentation and schematics throughout the design and build phases, maintaining accurate project records.
- Assisted in designing and assembling a custom motor driver, applying embedded systems and circuit design knowledge.

Student engineer – Kinetic Sculpture

Jul 2023 – Nov 2023

- Soldered wiring for lights and motors and conducted safety testing to ensure reliable system operation.
- Gathered raw data from motors to calibrate safety limits, ensuring operation within defined mechanical parameters.

Ball-balancing Beam PID Control System

Feb 2025 – Jul 2025

- Utilised MATLAB to simulate the physical characteristics of the ball-beam system, verifying gains for both open-loop and closed-loop system response.
- Implemented curve fitting libraries to reduce noise from digitisation of the analogue signal from the flexible potentiometer sensor.
- Successfully demonstrated closed-loop position control using a Teensy microcontroller, achieving stable regulation without motor encoder feedback.